

The data used in the 2024 sardine assessment

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Data to which the 2024 assessment of the South African sardine resource is to be conditioned are listed.

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The data used in the South African sardine assessments are not all raw data. Some data have already been subjected to analyses and refinements. These associated calculations are often done “behind the scenes” and their details are seldom recorded. This lack of record can result in a discontinuity in the method used to calculate data for subsequent assessments, particularly if assumptions made in the calculations are not documented and/or a new person becomes responsible for developing the data to be used for input to the assessment. This document serves to record the generation of the data used in the sardine assessment carried out in 2024 – using data up to the end of 2023 - from the raw data.

Information on Ageing

Age-length keys for the November surveys from 1984-1999 and for commercial catch for some months each year from 1984 to 1999 were derived by Michael Kerstan (De Oliveria 2003). Sardine age-length keys for the November surveys derived by Deon Durholtz are available by area (east and west of Cape Agulhas) for 1993, 1994, 1996, 2001 - 2004 and 2006 - 2009. Cynthia Mtengwane also compiled sardine age-length keys by area for the November surveys in 1993, 1994, 1996, 2001, 2003, 2004, 2006 - 2010 as well as selected monthly age-length keys for sardine commercial catch between 2004 and 2009. However, inconsistencies between these age-length keys derived by Kerstan and those from Durholtz and Mtengwane restricted the use of age-length keys from all readers in the assessment. In addition, Smith *et al.* (2011) recommended that these age-length keys not be used to calculate proportions-at-age for use in the assessments due to an inability to detect strong cohorts in the age data (which are known to be present through inference from the survey estimates of abundance). This may be due to problems with ageing and/or problems with the construction of the survey length-frequency data.

Sardine Commercial Data

Monthly length frequencies were constructed for the sardine landings using the method in Appendix A. From 1987 onwards, these have been split by area (north of Cape Point, Cape Point to Cape Agulhas and east of Cape Agulhas). For the two-area model, the catch tonnage and length frequencies by month west of Cape Agulhas are assumed to be equal to the combined catch tonnage for the north-west and south-west coasts.

Between 1987 and 2010, sardine landings were categorized as either directed (>50% sardine mass in landing) or bycatch by the scale monitor. The bycatch was recorded as being either caught with anchovy or round herring, with the allocation determined by the species which formed >50% mass in the landing.

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From 2011 onwards, the sardine landings have again been categorized as either directed >14cm (>50% sardine mass in landing) or bycatch by the scale monitor (de Goede 2010, van der Westhuizen 2010). The bycatch is now recorded as either ‘small’ (≤ 14 cm) sardine with directed >14cm sardine, or else ‘small’ (≤ 14 cm) or ‘large’ (>14cm) bycatch resulting from directed anchovy or round herring fishing. As fish of a similar size tend to shoal together, the assumption is made for this assessment, that the ‘small’ sardine bycatch is primarily bycatch with anchovy and the time series is comparable with the 1987-2010 time series of sardine bycatch with anchovy. Anchovy is seldom¹ landed with adult sardine and/or round herring. The ‘large’ sardine bycatch is assumed to be primarily bycatch with round herring and the time series is assumed comparable with the 1987-2010 time series of bycatch with round herring.

Monthly directed sardine catch numbers are available for 1981 to 1983 while monthly sardine catch tonnage and length frequencies are also available for 1984 to 1986 (De Oliveira pers. comm.). The following assumptions are made concerning the October 1983 to December 1986 data:

- All catches are taken to correspond to the time series of directed >14cm sardine catch. Any ≤ 14 cm sardine would thus be modelled as bycatch in the directed fishery.
- All catches are assumed to be taken west of Cape Agulhas. This is considered reasonable given the fishery historically operated off the west coast and the eastward shift in sardine distribution was only observed in later years.
- For the three area model, the quarterly catch tonnages and quarterly length frequencies west of Cape Agulhas are divided between those taken north and south of Cape Point according to the average annual proportion of catch tonnage in these two areas from 1987 to 1991.

Given the uncertainty associated with these data, the model is not fit to proportion-at-length in quarter 4 of 1983 and the model fit to commercial proportion-at-length between 1984 and 1986 is downweighted.

The sardine bycatch with anchovy (or ‘small’ sardine bycatch) is used separately in the assessment to the directed sardine catch and sardine bycatch with round herring. Quarterly data used in the assessments are taken over the months January to March, April to June, July to September, and October to December.

Directed sardine and >14cm sardine bycatch with round herring

The directed sardine and large sardine bycatch length frequencies are available by area, month and year, i.e. the number of fish in length class l in area a during month m of year y , $N_{y,l,m,a}$, from January 1987 to December 2023, together with the observed tonnage in each area by month and year, $ObsT_{y,m,a}$. The monthly catch tonnage is shown by area in Tables 1a-1c.

Expected mass by length class and month is calculated as: $EM_{y,l,m,a} = 0.0071118 \times l_{mid}^{3.181} \times N_{y,l,m,a}$, where l_{mid} is the mid-point of the length class considered and mass is in grams and length in millimetres (OLSPS 2023). The “raised” length frequency used in the assessment, is then calculated as follows: $\hat{N}_{y,m,l,a} = N_{y,m,l,a} \times \frac{ObsT_{y,m,a}}{\sum_i EM_{y,l,m,a}}$. For two area model, the raised length frequency for west of Cape Agulhas is then given by $\hat{N}_{y,m,l,a=2} + \hat{N}_{y,m,l,a=3}$, where $a = 2$ denotes Cape Point to Cape Agulhas and $a = 3$ denotes north of Cape Point.

¹ Occasionally in the area from Cape Point to Cape Agulhas.

Between October 1983 to December 1986 no directed sardine or large sardine bycatch was assumed landed east of Cape Agulhas (see above section). The raised length frequency for west of Cape Agulhas is calculated using the original data, i.e.:

$$\hat{N}_{y,m,l,2+3} = N_{y,m,l,2+3} \times \frac{ObsT_{y,m,2+3}}{\sum_l EM_{y,l,m,2+3}}. \text{ For the three area model it is assumed (see above section), } \hat{N}_{y,m,l,2} = \hat{N}_{y,m,l,2+3} \times \frac{1}{5} \sum_{y=1987}^{1991} \frac{ObsT_{y,m,2}}{ObsT_{y,m,2+3}} \text{ and } \hat{N}_{y,m,l,3} = \hat{N}_{y,m,l,2+3} \times \frac{1}{5} \sum_{y=1987}^{1991} \frac{ObsT_{y,m,3}}{ObsT_{y,m,2+3}}.$$

Sardine bycatch with anchovy

The small sardine bycatch length frequencies are available by area, month and year from 1987 to 2023 are stored in, together with the observed tonnage in each month and year, $ObsT_{y,m,a}$. The monthly catch tonnages are given in Table 1d-1e. In the absence of data, small sardine bycatch between 1983 and 1986 is assumed to be zero.

Small amounts of sardine bycatch with anchovy totalling 28.1t between 1989 and 2015, 54.0t between 2016 and 2019, 112.5t between 2020 and 2023 were recorded east of Cape Agulhas. While previous assessments assumed these small bycatches were taken west of Cape Agulhas with the anchovy bycatch between Cape Point and Cape Agulhas, sensitivity to this assumption will now be tested given the increasing trend in these bycatches. As there was no length frequency recorded with the 0.07t, 0.175t and 0.16t of small sardine bycatch east of Cape Agulhas in January 1989, April 2015 and April 2017, respectively, this small catch was ignored.

These sardine bycatch with anchovy data are split between juvenile (0-year-old) and adult catch as follows. Let $N_{y,m,l,a}$ denote the number of sardine in length class l landed as bycatch with anchovy in month m of year y and area a . The ‘raised’ length frequency is calculated in the same manner as for the directed catches above, i.e. $\hat{N}_{y,m,l,a} = N_{y,m,l,a} \times \frac{ObsT_{y,m,a}}{\sum_l EM_{y,l,m,a}}$. Juvenile sardine bycatch with anchovy landed in month m of year y is taken to be all sardine below a given cut-off length, i.e. $C_{y,m,0,a} = \sum_{l=lmin}^{lcut(y,m)} \hat{N}_{y,m,l,a}$. The remaining sardine bycatch with anchovy is assumed to be 1 year olds, i.e. $C_{y,m,1} = \sum_{l=lcut(y,m)}^{lmax} \hat{N}_{y,m,l,a}$. For the two area model, the sardine bycatch west of Cape Agulhas is then given by $C_{y,m,0,a=2} + C_{y,m,0,a=3}$, where $a = 2$ denotes Cape Point to Cape Agulhas and $a = 3$ denotes north of Cape Point.

The cut-off length, $lcut(y, m)$, taken to apply to May and June was set at that used during the recruit survey. This cut-off was originally determined from analyses of age-growth curves (1984-1995) but, because of the lack of reliable age information since the mid-1990s, based on a modal progression analysis (1996-2004). The cut-off lengths have recently been revised to the length at 20% maturity for years in which macroscopic gonad staging information is available (Coetzee and Merkle 2007, Yemane *et al.* 2024, given in Table 2). The cut-off length was decreased for months from May back to November, and increased from June through to October. This is in line with de Moor *et al.* (2020) and the November survey length frequencies, both back from May to November of the previous year and forward to November of the current year were considered (Appendix C). A faster growth rate was assumed in the earlier months:

Month	Change in recruit cut-off length from that used for the recruit survey in year y
October-December	-6cm
January-February	-3cm
March-April	-1cm
May-June	0
July-August	+1cm
September	+1.5cm

In 1987 and 2022 the recruit survey began in July. Thus for these years the cut-off length used for May-June is 1cm less than that of the recruit survey, i.e. 10.5cm and 11.5cm, respectively. A cut-off length of 13.5cm was assumed for May/June 1984, corresponding approximately to the average of the 1985 and 1986 cut-off lengths. For May/Jun 2018, a cut-off length of 11cm was assumed, corresponding to the average of the 2017 and 2019 cut-off lengths.

Recruit catch prior to the survey

As catch is modelled quarterly, the sardine 0-year-old catch prior to the recruit survey is required only from 1 April to the day before the survey commenced. This is calculated from the ‘raised’ length frequencies of landings between 1 April and the day before the commencement of the survey (totalled over all catches and bycatches). The number of 0-year old sardine landed from 1 April to the day before the survey is taken to be $C_{y,a}^{BS} = \sum_{m=4}^{m<pth(y)} \sum_{l=lmin}^{l<cut(y,6)} \dot{N}_{y,m,l,a} + \sum_{l=lmin}^{l<cut(y,6)} \dot{N}_{part,y,pth(y),l,a}$, where $pth(y)$ denotes the month in which the recruit survey began in year y and $\dot{N}_{part,y,pth(y),l,a}$ denotes the ‘raised’ length frequency from landings up to the day before the survey began during the month in which the survey began. ‘Raised’ length frequencies for these part-months are calculated separately for small sardine bycatches and for directed sardine and large sardine bycatches (as is done for all other months, see above). The cut-off lengths used to calculate the recruit survey biomass, also used to calculate the recruit catch in May and June (see above) were applied.

As the length frequencies are only available west and east of Cape Agulhas for the part-month in which the survey began, and not separately for the north-west and south-west coasts, the ‘raised’ length frequency is calculated for the west coast from the length frequency provided for the west coast for all years. For the three area model it is assumed that the raise length frequencies for the part-month in which the survey began are in proportion to the observed catch tonnage by area for that full month, i.e., $\dot{N}_{part,y,pth(y),l,2} = \dot{N}_{part,y,pth(y),l,2+3} \times \frac{ObsT_{y,pth(y),2}}{ObsT_{y,pth(y),2+3}}$ and $\dot{N}_{part,y,pth(y),l,3} = \dot{N}_{part,y,pth(y),l,2+3} \times \frac{ObsT_{y,pth(y),3}}{ObsT_{y,pth(y),2+3}}$.

In 1985 and 1986 length frequencies for the part-months in which the recruit survey began were not directly available as inspector data (which include samples for species split) were not available in 1985 and 1986. However, daily skippers’ estimates of tonnage landed were available for these years. Although the total tonnage landed in May 1985 and June 1986 was estimated by the skippers to be different to that arising from the source data, it was assumed that the proportion of catch taken before the survey compared to the whole month was the same between the skippers’ estimates and the source data. Thus “raised” length frequencies for 1-19 May 1985 and 1-9 June 1986 were calculated by reducing the catch tonnage proportionally as follows: $\dot{N}_{y,m,l,2+3} = N_{y,m,l,2+3} \times \frac{ObsTy,m,a}{\sum_i EM_{y,l,m,a}} \times \frac{SkipperT_{partmonth}}{SkipperT_{fullmonth}}$, using the data in the below table. For the three area model, the raise length frequencies for the north-west and south-west coasts are calculated in the same manner as for other years (see above).

	Catch for the month (tons)	Skipper estimated catch for the month (tons)	Skipper estimated catch prior to the survey (tons)
May 1985	3274	479	205
June 1986	4042	970	609

November Survey Data

The time series of total biomass estimates and associated CVs from the hydroacoustic surveys in November each year, corresponding to the standard survey area between Hondeklip Bay and Port Alfred, are given in Table 3. These data are stored in *SurveyData.xls*. The early part of this time series was updated by de Moor *et al.* (2008) so that the full time series corresponds to uncapped survey biomass determined with a new target strength formula (Coetzee *et al.* 2008) and corrected for attenuation. Coetzee *et al.* (2024a) recently revised the full time series to standardise strata across years, to extrapolate the biomass to the full survey area in years for which the survey did not cover the full area and to include the new length-weight regression $w_l = 0.0071118 \times l^{3.181}$ (OLSPS 2023). Note that the regression used to convert capped to uncapped data has not been updated from that used by de Moor *et al.* (2008) with the revised and extended timeseries.

Length frequencies are also available for the hydroacoustic surveys.

A survey in February/March 2024 was conducted after the November 2023 survey was cancelled (Coetzee *et al.* 2024b). The assessment fits model predicted biomass and associated proportions-at-length from 15th February 2024 (rather than 15th November 2023) to these February/March 2024 data (see de Moor 2024 for further details).

Recruit Survey Data

The time series of recruit biomass and associated CVs from the May/June recruit surveys is given in Table 4. The recruit numbers at the time of the survey were calculated by summing the number of fish smaller than a cut-off length in the weighted length frequency (Appendix B²). In the assessments, the recruit numbers are used together with the CVs on recruit biomass. To align with the two-stock hypothesis, these data are now split at Cape Agulhas instead of Cape Infanta³. These survey data are stored in *SurveyData.xls*. The early part of this time series was updated by de Moor *et al.* (2008) and revised again in 2011 so that the full time series corresponds to uncapped recruitment determined with a new target strength formula (Coetzee *et al.* 2008) and corrected for attenuation. Coetzee *et al.* (2024a) recently revised the full time series to standardise strata across years, to extrapolate the biomass and recruit numbers to the survey area up to Cape Infanta in years for which the survey did not extend as far east as Cape Infanta and to include the new length-weight regression $w_l = 0.0081479 \times l^{3.121}$ (OLSPS 2023). Note that the regression used to convert capped to uncapped data has not been updated from that used by de Moor *et al.* (2008) with the revised timeseries.

Parasite Data

The time series of infection prevalence of the “tetracotyle” type digenean endoparasite by length corresponding to Novembers 2010 to 2020 and 2022 is given in Table 5. These data are a combination of samples from both the November surveys (e.g. van

² An alternative method of deriving the WLF (converting the acoustic biomass to numbers at length) has been implemented for sardine in November surveys. This does away with the weighting of each trawl LF by the acoustic density assigned to it and rather uses a L/W regression to calculate numbers at length based on proportion at length from the pooled stratum sample LFs (Coetzee and Merkle 2011). This adjustment has not yet been implemented for sardine recruit surveys and will result in revised number of recruits and recruit mean weight. The extent of such change is unknown at this stage.

³ The area west of Cape Infanta has historically been the standard recruit survey area and is used for an estimate of anchovy recruitment.

der Lingen and Petersen 2020) and commercial catches in October to December (e.g. Weston *et al.* 2015). van der Lingen (2021) demonstrated that the prevalence-at-length in these two sets of data could be combined thereby increasing the coverage of, particularly, larger fish off the south coast in more recent years.

These prevalence data represent the proportion of sardine-by-length that are infected with the parasite. The prevalence data collected from west of Cape Agulhas (20°E) are not separated between 'northwest' and 'southwest' coasts, even in a three-area model (van der Lingen 2023). Model predicted prevalence-at-length for sardine distributed off the south coast are fit to samples collected between 22°E (roughly Mossel Bay) and 30°E (roughly Port St Johns) to exclude age-1 individuals in the hypothesized mixing zone (20°-22°E) that Dunn *et al.* (2015) suggested may be western component fish under the previous stock structure hypothesis. An alternative time series of south coast prevalence based on samples collected between Cape Agulhas and 30°E is used for a model sensitivity test.

Alternative information on the intensity of parasite infection, i.e. numbers of parasite per infected fish, is also available and may be included in later analyses.

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Table 1a. The monthly sardine commercial catch tonnage (in thousands of tons) landed as directed catch (1987⁴-2023) or bycatch with the round herring fishery (1987-2010) or 'large' sardine bycatch (2011-2023), west of Cape Point.

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1987	2.480	4.815	4.642	4.786	1.510	1.398	0.266	0.218	0.066	0.000	0.000	0.000
1988	0.926	3.255	0.714	4.274	2.951	3.951	2.490	1.323	0.353	0.208	0.911	0.657
1989	0.797	0.970	5.750	1.988	2.010	2.716	1.908	0.670	0.136	0.016	0.045	0.072
1990	2.574	4.359	6.081	4.882	7.261	4.405	2.630	5.074	1.332	0.461	0.051	0.306
1991	2.340	4.386	2.206	3.780	3.674	0.543	0.080	0.004	0.000	0.000	0.000	0.008
1992	0.315	4.376	3.991	0.093	0.123	2.376	0.265	1.462	0.095	0.004	0.041	0.802
1993	4.534	6.663	4.899	0.993	4.015	3.065	0.456	0.493	0.560	0.600	1.115	1.652
1994	0.346	3.777	6.603	6.090	14.327	1.656	1.932	3.865	3.222	4.762	1.028	0.369
1995	2.474	4.859	9.744	3.924	4.889	1.595	2.885	11.694	7.287	10.629	4.958	0.114
1996	2.787	8.260	7.963	6.472	8.096	6.855	4.801	6.465	7.696	8.023	9.654	1.159
1997	0.847	6.909	8.464	10.538	12.001	5.674	10.892	8.929	13.472	7.553	2.872	0.212
1998	1.366	6.899	12.283	4.030	7.238	10.918	8.336	9.667	9.615	12.321	4.257	0.664
1999	2.013	0.119	3.906	1.443	4.159	3.639	11.871	12.832	13.009	4.400	18.095	9.691
2000	0.000	2.274	6.286	6.846	0.898	4.824	1.696	5.941	6.629	16.577	12.494	1.341
2001	2.029	6.727	3.435	2.021	0.718	1.395	0.880	1.480	2.197	1.660	20.772	25.471
2002	0.000	10.120	6.574	18.198	1.242	0.631	0.975	3.403	6.031	9.044	24.772	27.043
2003	3.057	23.769	16.794	3.993	1.252	1.019	0.451	0.000	2.112	11.069	25.232	8.155
2004	0.000	0.735	2.817	0.430	1.432	0.651	1.583	0.520	21.808	3.355	3.061	0.000
2005	0.000	0.782	0.307	0.225	0.159	0.029	0.000	0.086	0.000	0.171	0.181	0.000
2006	0.000	0.208	0.864	0.216	0.381	0.884	0.115	0.009	0.415	10.328	0.519	1.991
2007	0.000	0.098	0.000	3.141	4.989	0.019	2.116	4.775	5.250	0.343	2.294	1.125
2008	0.000	0.175	0.490	0.018	0.122	0.528	0.386	0.243	0.000	7.055	8.946	2.577
2009	0.049	2.215	1.849	0.291	0.143	1.883	0.833	0.698	0.063	0.227	0.000	0.000
2010	0.805	3.683	2.426	2.155	0.734	0.208	0.235	0.296	0.000	0.000	8.125	0.673
2011	0.287	3.592	6.941	2.307	1.632	0.884	7.743	1.798	0.504	0.555	0.001	0.000
2012	2.109	2.855	1.043	0.442	0.051	0.003	0.126	0.122	1.228	0.918	0.567	0.000
2013	1.837	0.213	0.000	0.489	0.525	0.184	0.000	0.000	0.000	0.000	0.421	0.000
2014	0.057	0.048	0.220	1.260	0.719	0.031	0.014	1.660	0.303	0.000	0.000	0.000
2015	0.000	0.033	0.181	0.574	1.082	0.122	0.000	0.000	0.000	2.012	2.326	0.000
2016	0.088	0.738	1.561	3.874	1.229	0.741	0.018	0.060	0.022	0.011	0.724	0.000
2017	0.010	0.122	0.083	0.154	0.371	0.000	0.000	0.000	0.000	0.000	0.206	0.040
2018	0.121	0.197	1.309	0.005	0.134	0.003	0.000	0.000	0.005	0.232	0.232	0.129
2019	0.000	0.065	0.010	0.147	0.021	0.034	0.007	0.007	0.114	0.000	0.000	0.000
2020	0.002	0.259	1.006	0.437	0.061	0.245	0.110	0.073	0.001	0.273	0.237	0.000
2021	0.000	0.748	0.331	2.102	0.524	0.166	0.052	0.083	0.007	0.037	0.000	0.000
2022	0.070	0.000	0.428	0.485	0.070	0.031	0.057	1.065	0.218	0.000	0.145	0.000
2023	0.736	0.504	0.718	0.159	0.941	1.604	2.014	0.179	0.109	1.020	0.000	0.000
2024	0.023	0.173	6.480	-	-	-	-	-	-	-	-	-

⁴ See main text for assumptions regarding directed catches prior to 1987.

Table 1b. The monthly sardine commercial catch tonnage (in thousands of tons) landed as directed catch (1983⁴-2023) or bycatch with the round herring fishery (1987-2010) or 'large' sardine bycatch (2011-2023), between Cape Point and Cape Agulhas.

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1983										0.277	0.072	0.083
1984	1.980	6.802	4.975	6.520	5.114	1.361	0.010	0.000	0.000	0.261	0.131	0.000
1985	3.641	5.715	6.198	4.255	3.274	5.640	1.964	0.011	0.014	0.000	0.000	0.000
1986	1.310	7.319	8.638	3.539	2.714	4.042	2.855	0.162	0.060	0.000	0.000	0.000
1987	1.195	1.507	2.370	0.852	0.341	0.000	0.258	0.000	0.000	0.000	0.000	0.000
1988	0.898	2.057	2.026	1.618	0.953	0.208	0.134	0.000	0.000	0.000	0.001	0.000
1989	0.576	1.578	1.713	2.351	0.629	0.263	0.030	0.104	0.041	0.020	0.130	0.000
1990	0.444	1.655	1.595	1.687	2.078	0.419	0.958	0.075	0.383	0.233	0.294	0.122
1991	0.186	1.742	1.811	2.379	3.776	5.009	5.619	3.989	1.586	1.098	0.124	0.180
1992	0.467	0.771	1.604	2.238	1.845	4.679	2.612	3.885	5.956	1.083	0.250	0.139
1993	0.103	1.205	1.612	3.308	2.437	2.227	0.572	0.497	0.348	0.566	0.191	0.057
1994	1.346	2.487	4.772	1.790	2.051	4.569	4.764	3.432	1.440	0.443	0.196	0.008
1995	0.228	1.176	1.389	2.331	8.951	4.836	2.963	3.251	1.026	2.205	0.392	0.222
1996	0.105	0.763	1.486	1.274	2.191	0.881	0.849	1.125	1.138	2.317	1.564	0.308
1997	0.364	1.537	2.366	1.770	1.969	1.096	2.867	2.949	4.380	0.101	0.292	0.158
1998	1.018	1.520	1.983	2.214	1.252	2.252	4.887	9.049	1.688	2.020	0.189	0.151
1999	0.207	0.106	1.290	3.989	8.751	4.751	1.834	1.969	1.937	1.835	4.686	0.763
2000	0.000	0.184	1.510	3.966	12.051	12.089	9.430	6.473	3.707	2.822	3.439	0.455
2001	0.251	3.961	13.772	11.308	11.995	9.813	4.992	7.017	2.130	23.870	4.967	3.457
2002	0.106	2.197	8.236	8.519	10.921	7.562	7.193	9.908	16.783	16.297	22.880	2.486
2003	0.838	1.539	12.331	17.240	13.498	11.120	5.754	1.838	1.565	11.900	34.004	9.888
2004	8.484	39.911	28.890	17.069	29.342	17.806	13.680	3.098	3.282	15.327	57.611	19.235
2005	0.211	19.073	28.983	18.047	0.850	0.129	1.118	0.044	0.067	4.097	9.967	1.410
2006	1.123	0.699	18.337	5.470	0.212	0.177	0.099	0.295	11.493	8.681	15.109	5.354
2007	3.474	7.405	5.919	2.639	2.029	1.648	1.486	0.102	1.365	3.556	0.556	0.050
2008	0.000	0.593	7.510	7.441	1.333	3.136	0.793	0.953	0.000	0.000	0.066	0.336
2009	0.001	6.837	16.046	11.920	7.420	3.153	2.359	1.213	0.000	0.016	0.161	0.003
2010	0.000	3.735	11.395	6.965	8.527	6.127	6.539	2.713	2.184	0.037	0.795	0.000
2011	0.341	4.079	8.614	5.336	4.567	3.115	4.199	4.818	6.160	2.335	0.125	0.026
2012	2.928	12.005	12.773	10.438	9.020	6.407	0.923	0.728	1.778	3.924	5.148	0.000
2013	0.000	12.047	12.554	10.947	6.380	0.963	0.000	0.000	0.220	2.504	3.376	0.897
2014	5.884	12.138	12.822	7.933	0.228	0.000	0.091	1.229	1.599	4.486	4.534	1.582
2015	1.003	9.504	10.850	3.888	0.000	4.275	1.532	2.311	0.564	7.911	12.150	1.526
2016	0.000	12.144	9.403	0.818	1.737	1.971	0.016	0.105	2.466	4.845	2.773	0.541
2017	0.254	0.466	8.141	1.753	0.778	0.008	0.006	0.007	0.831	1.138	7.975	4.787
2018	0.003	3.716	7.109	2.854	0.099	0.012	0.090	0.016	0.006	5.424	2.828	1.845
2019	0.001	0.021	0.137	0.117	0.077	0.041	0.000	0.000	0.168	0.065	0.171	0.286
2020	0.069	0.284	0.774	1.151	5.110	1.309	0.000	0.000	0.002	1.526	0.096	0.000
2021	0.003	0.076	0.114	1.742	0.795	1.791	0.000	0.001	0.000	0.001	0.069	0.000
2022	0.000	0.396	0.750	0.001	0.000	0.136	0.221	0.002	0.000	0.000	3.598	2.254
2023	0.000	1.543	2.339	2.663	1.875	0.243	0.000	0.000	0.000	0.180	1.088	0.180
2024	0.000	1.252	0.946	-	-	-	-	-	-	-	-	-

Table 1c. The monthly sardine commercial catch tonnage (in thousands of tons) landed as directed catch (1989-2023) or bycatch with the round herring fishery (1989-2010) or 'large' sardine bycatch (2011-2023), east of Cape Agulhas. There was no catch east of Cape Agulhas prior to 1989.

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1989	0.000	0.000	0.167	0.000	0.000	0.000	0.000	0.004	0.000	0.000	0.047	0.000
1990	0.011	0.031	0.153	0.061	0.046	0.031	0.059	0.014	0.000	0.000	0.057	0.016
1991	0.010	0.224	0.114	0.158	0.272	0.074	0.000	0.000	0.000	0.230	0.134	0.164
1992	0.039	0.155	0.544	0.387	0.338	0.201	0.013	0.056	0.126	0.352	0.205	0.051
1993	0.097	0.234	0.378	0.318	0.227	0.196	0.005	0.152	0.161	0.119	0.142	0.270
1994	0.011	0.633	0.270	0.315	0.561	0.607	0.534	0.481	0.144	0.395	0.072	0.345
1995	0.365	0.743	0.605	0.062	0.481	0.159	0.309	0.135	0.257	0.837	0.594	0.395
1996	0.064	0.533	0.456	0.400	1.073	0.731	0.625	0.539	0.672	0.398	1.136	0.915
1997	0.093	0.290	0.741	0.362	0.640	0.369	1.234	0.134	0.105	0.298	0.000	0.000
1998	0.012	0.000	0.536	0.612	0.972	1.156	0.554	0.069	0.168	0.016	0.100	0.000
1999	0.708	0.061	0.413	0.692	0.817	0.943	0.255	0.408	0.457	0.709	1.006	0.623
2000	0.000	0.271	0.541	0.754	1.444	1.133	0.138	0.688	0.357	0.172	0.505	0.044
2001	0.135	0.304	0.537	0.497	0.657	0.992	1.253	1.798	2.178	1.481	1.152	0.296
2002	0.000	0.885	0.671	0.678	2.493	2.880	4.275	4.873	3.314	3.051	2.712	1.419
2003	0.586	2.005	2.172	2.669	6.255	7.391	9.603	6.849	9.180	6.531	6.066	1.693
2004	0.534	1.660	2.543	4.306	7.630	10.285	10.250	15.521	9.307	9.738	4.287	1.394
2005	0.468	4.889	5.332	10.422	19.516	24.672	25.615	18.544	18.181	9.052	16.047	2.232
2006	0.947	6.454	10.630	12.736	28.192	25.894	17.695	8.775	3.450	3.823	3.469	3.114
2007	0.441	6.538	10.763	12.977	16.470	15.113	7.227	4.603	3.252	0.160	2.033	1.608
2008	0.344	2.088	3.175	13.837	8.529	3.685	7.192	2.254	0.236	1.055	1.055	0.567
2009	0.671	2.725	4.318	6.829	7.009	4.400	3.328	0.374	0.932	1.267	0.876	1.412
2010	0.814	2.443	3.156	2.836	3.460	3.256	3.030	3.262	2.607	0.292	0.032	0.905
2011	0.419	3.115	3.551	2.018	4.591	3.571	3.719	4.794	2.347	0.905	1.330	0.434
2012	1.048	2.582	4.278	4.027	5.513	4.092	1.958	0.863	0.807	0.215	0.312	0.437
2013	0.476	1.900	2.732	4.032	4.811	6.063	5.114	5.657	1.739	2.378	0.096	0.210
2014	0.088	0.669	3.832	3.782	6.940	6.277	9.034	1.683	1.696	0.439	0.000	0.000
2015	0.000	0.003	0.347	3.988	7.164	3.915	3.333	2.246	0.172	0.000	0.000	0.000
2016	0.000	0.036	0.004	1.612	10.258	3.997	3.764	2.825	0.000	0.062	0.000	0.000
2017	0.041	0.119	0.318	0.001	1.591	0.807	0.067	0.000	0.058	0.534	0.784	0.775
2018	0.033	0.294	0.767	0.806	2.492	1.040	1.193	0.029	0.550	1.193	1.518	1.385
2019	0.000	0.000	0.000	0.032	0.000	0.034	0.283	0.130	0.114	0.000	0.546	0.230
2020	0.000	0.000	0.044	0.505	0.495	0.336	0.053	0.025	0.071	0.411	1.485	1.454
2021	0.000	0.570	1.758	2.192	3.511	2.488	2.250	0.337	0.157	1.192	1.831	1.418
2022	1.384	1.614	4.827	3.421	2.605	0.941	0.309	0.407	0.189	1.370	1.341	0.095
2023	0.043	1.078	0.678	1.295	3.008	0.278	2.528	2.635	1.499	0.258	2.894	0.750
2024	0.719	1.472	1.561	-	-	-	-	-	-	-	-	-

Table 1d. The monthly sardine commercial catch tonnage (in thousands of tons) landed as bycatch with the anchovy fishery (1987-2010) or ‘small’ sardine bycatch (2011-2023), north of Cape Point.

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1987	0.005	0.163	0.225	1.360	0.321	1.462	1.517	1.407	0.206	0.000	0.000	0.000
1988	0.022	0.233	0.100	0.020	1.009	2.381	0.520	0.724	0.154	0.689	0.235	0.000
1989	0.023	1.989	0.935	1.749	2.685	1.565	0.292	0.000	0.000	0.000	0.000	0.000
1990	0.019	0.175	0.467	0.903	2.030	3.797	0.012	0.000	0.000	0.000	0.000	0.000
1991	0.001	0.072	1.365	2.778	0.485	1.935	0.026	0.004	0.000	0.000	0.000	0.000
1992	0.079	0.388	0.267	1.775	1.668	2.546	2.231	2.724	0.209	0.008	0.000	0.000
1993	0.070	0.156	0.397	1.397	1.298	0.204	0.619	1.547	0.559	0.163	0.000	0.000
1994	0.168	1.462	0.471	1.327	4.281	1.936	0.039	3.460	0.032	0.000	0.000	0.000
1995	0.046	0.022	1.025	0.686	1.822	4.183	5.076	6.123	0.388	1.970	0.535	0.000
1996	1.015	1.789	0.663	0.566	1.747	1.785	0.000	0.000	0.000	0.004	0.000	0.000
1997	0.073	0.000	0.005	0.002	0.237	0.267	0.980	0.664	2.434	0.863	0.000	0.000
1998	0.028	1.118	0.122	1.720	3.552	4.468	0.960	0.180	0.574	0.005	0.000	0.000
1999	0.000	0.000	0.318	0.381	1.352	2.071	0.490	0.722	1.331	0.425	0.000	0.000
2000	0.000	0.000	1.368	1.613	1.552	0.949	0.322	0.013	0.022	0.001	0.000	0.000
2001	0.001	0.231	0.154	0.980	2.177	2.548	1.076	3.431	1.145	1.689	0.046	0.028
2002	0.040	0.185	0.000	0.350	0.327	1.836	1.297	5.537	2.475	0.000	0.000	0.009
2003	0.000	0.000	0.182	1.845	2.103	4.282	1.094	0.118	0.269	0.462	0.128	0.000
2004	0.000	0.017	0.002	0.956	3.209	0.469	0.700	0.583	0.174	0.000	0.000	0.000
2005	0.000	0.072	0.959	0.863	1.480	0.322	0.388	0.242	0.520	0.257	0.051	0.000
2006	0.000	0.000	0.005	0.321	0.539	2.082	2.646	0.884	1.629	0.008	0.000	0.000
2007	0.000	0.001	0.020	0.720	1.763	0.268	0.201	0.283	0.123	0.120	0.000	0.000
2008	0.000	0.000	0.000	0.123	1.420	0.746	0.259	0.268	0.130	0.042	0.015	0.004
2009	0.000	0.041	0.046	0.709	0.220	0.072	0.204	0.370	0.059	0.019	0.000	0.000
2010	0.088	0.155	1.707	2.096	2.442	5.345	4.023	0.731	0.036	0.000	0.000	0.000
2011	0.008	0.046	0.146	1.409	3.082	1.257	3.640	1.087	0.000	0.000	0.000	0.000
2012	0.553	0.913	0.462	0.922	2.774	0.650	0.237	0.171	0.161	0.461	0.000	0.000
2013	0.053	0.000	0.000	0.517	2.007	0.628	0.006	0.000	0.001	0.000	0.000	0.000
2014	0.000	1.039	1.242	1.951	1.550	0.015	0.026	0.109	0.023	0.006	0.000	0.000
2015	0.000	1.661	3.595	2.289	4.626	0.196	0.039	0.007	0.003	0.000	0.000	0.000
2016	0.001	0.724	0.046	3.739	3.539	1.769	0.093	0.021	0.080	0.000	0.000	0.000
2017	0.018	0.040	0.056	1.405	1.567	0.266	0.304	0.029	0.003	0.000	0.004	0.000
2018	0.003	0.022	0.029	0.063	0.521	0.077	0.085	0.114	0.026	0.001	0.002	0.002
2019	0.043	0.078	0.133	0.661	0.412	0.216	0.148	0.049	0.233	0.012	0.000	0.000
2020	0.000	0.004	0.105	2.478	1.640	0.403	0.944	0.300	0.067	0.018	0.026	0.000
2021	0.000	0.168	0.053	0.828	1.591	1.177	0.057	1.307	0.023	0.000	0.000	0.000
2022	0.000	0.000	0.134	0.837	1.513	0.694	0.060	0.035	0.650	0.000	0.001	0.000
2023	0.010	0.206	0.395	1.048	2.770	0.296	0.444	0.004	0.054	0.040	0.000	0.000
2024	0.340	0.017	0.250	-	-	-	-	-	-	-	-	-

Table 1e. The monthly sardine commercial catch tonnage (in thousands of tons) landed as bycatch with the anchovy fishery (1987-2010) or ‘small’ sardine bycatch (2011-2023), between Cape Point and Cape Agulhas. These data include the small amounts of sardine landed east of Cape Agulhas.

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1987	0.013	0.024	0.055	0.055	0.008	0.000	0.005	0.000	0.000	0.000	0.000	0.000
1988	0.010	0.058	0.015	0.038	0.207	0.010	0.000	0.000	0.000	0.000	0.000	0.000
1989	0.112	0.154	0.035	0.033	0.304	0.011	0.106	0.000	0.000	0.000	0.000	0.000
1990	0.000	0.018	0.011	0.109	0.043	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1991	0.010	0.002	0.108	0.000	0.034	0.239	0.003	0.001	0.000	0.000	0.000	0.000
1992	0.062	0.113	0.198	0.682	0.000	0.019	0.050	0.044	0.067	0.000	0.000	0.000
1993	0.000	0.023	0.103	0.000	0.078	0.000	0.001	0.004	0.000	0.000	0.000	0.000
1994	0.117	0.510	1.212	0.032	0.166	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1995	0.000	0.005	0.000	0.049	0.068	0.123	0.000	0.009	0.059	0.000	0.000	0.000
1996	0.000	0.142	0.026	0.058	0.099	0.175	0.007	0.000	0.000	0.000	0.000	0.000
1997	0.000	0.006	0.000	0.000	0.006	0.000	0.489	0.071	0.792	0.000	0.000	0.000
1998	0.000	0.000	0.021	0.042	0.122	0.024	0.000	0.004	0.124	0.257	0.000	0.000
1999	0.000	0.000	0.000	0.000	0.012	0.217	0.000	0.008	0.063	0.057	0.089	0.000
2000	0.000	0.000	0.035	0.185	0.345	0.197	0.288	0.304	0.008	0.020	0.000	0.000
2001	0.000	0.014	0.090	0.001	0.081	0.075	0.021	0.000	0.146	0.000	0.000	0.000
2002	0.000	0.000	0.000	0.002	0.075	0.000	0.000	0.145	0.234	0.000	0.000	0.000
2003	0.000	0.000	0.000	0.000	0.033	0.007	0.036	0.000	0.011	0.000	0.002	0.000
2004	0.000	0.000	0.000	0.000	0.089	0.006	0.006	0.021	0.012	0.000	0.003	0.000
2005	0.000	0.000	0.036	0.416	0.027	0.062	0.006	0.018	0.000	0.009	0.080	0.000
2006	0.000	0.000	0.136	0.031	0.159	0.221	0.118	0.095	0.189	0.058	0.006	0.000
2007	0.000	0.002	0.041	0.004	0.209	0.098	0.001	0.008	0.000	0.071	0.000	0.004
2008	0.000	0.042	0.156	0.380	0.042	0.010	0.030	0.222	0.007	0.049	0.258	0.000
2009	0.000	0.025	0.135	0.067	0.162	0.255	0.156	0.194	0.000	0.062	0.010	0.000
2010	0.000	0.032	0.150	0.028	0.070	0.011	0.144	0.867	0.000	0.046	0.015	0.000
2011	0.000	0.019	0.016	0.115	0.290	0.000	0.147	0.128	0.000	0.000	0.000	0.000
2012	0.000	0.000	0.138	0.026	0.082	0.004	0.016	0.018	0.055	0.034	0.017	0.000
2013	0.000	0.000	0.000	0.109	0.004	0.005	0.000	0.000	0.004	0.000	0.054	0.000
2014	0.000	0.032	0.004	0.006	0.000	0.000	0.000	0.004	0.024	0.002	0.000	0.000
2015	0.000	0.016	0.627	0.045	0.007	0.217	0.018	0.001	0.000	0.002	0.001	0.000
2016	0.000	0.084	0.048	0.045	0.306	0.436	0.059	0.302	0.491	0.020	0.001	0.000
2017	0.000	0.173	0.120	0.005	0.294	0.008	0.025	0.025	0.007	0.144	0.208	0.013
2018	0.000	0.023	0.104	0.264	0.027	0.005	0.022	0.061	0.010	0.050	0.032	0.051
2019	0.000	0.070	0.076	0.024	0.082	0.002	0.009	0.008	0.078	0.015	0.000	0.000
2020	0.003	0.195	0.198	0.079	0.067	0.027	0.029	0.001	0.033	0.037	0.000	0.000
2021	0.002	0.034	0.015	0.038	0.042	0.108	0.003	0.011	0.026	0.000	0.013	0.000
2022	0.000	0.000	0.007	0.020	0.005	0.177	0.299	0.063	0.055	0.000	0.021	0.004
2023	0.000	0.005	0.006	0.008	0.119	0.008	0.000	0.002	0.002	0.051	0.026	0.001
2024	0.000	0.068	0.043	-	-	-	-	-	-	-	-	-

Table 2. The date of the commencement of the annual recruit survey (taken to be the date on which the first acoustic interval is logged and not the date on which the first station was occupied); the cut-off lengths used to estimate juvenile sardine from the recruit surveys.

Year	Date of commencement of survey	Time of the recruit survey after 1 May	Cut-off length (cm) for sardine juveniles in the survey
1985	20-May	0.613	<14.0
1986	10-June	1.300	<12.5
1987	20-July	2.613	<11.5
1988	27-June ⁵	1.867	<9.0
1989	08-June ⁶	1.233	<12.0
1990	22-June	1.700	<12.5
1991	07-May	0.194	<12.5
1992	13-May	0.387	<12.5
1993	21-May	0.645	<12.5
1994	05-May	0.129	<12.5
1995	10-June	1.300	<14.5
1996	05-June	1.133	<14.0
1997	17-May	0.516	<12.0
1998	20-May	0.613	<12.5
1999	10-May	0.290	<14.5
2000	15-May	0.452	<14.5
2001	05-May	0.129	<12.0
2002	05-May	0.129	<15.0
2003	14-May	0.419	<15.5
2004	08-May	0.226	<14.0
2005	13-May	0.387	<12.5
2006	19-May	0.581	<13.0
2007	18 May	0.548	<12.5
2008	21 May	0.645	<12.0
2009	15 May	0.452	<14.5
2010	27 May	0.839	<13.5
2011	27 May	0.839	<15.0
2012	16 June	1.500	<13.5
2013	23 May	0.710	<11.5
2014	10 May	0.290	<13.5
2015	22 May	0.677	<13.5
2016	8 June	1.233	<14.0
2017	12 June	1.367	<10.5
-	-	⁷	⁸
2019	7 June ⁹	1.200	<11.5
2020	17 June	1.533	<11.5
2021	9 June	1.267	<11.0
2022	2 July	2.032	<12.5
2023	25 May	0.774	<12.0

⁵ The first station was on 27th June 1988, although the first acoustic interval was only logged after midnight, i.e. on 28th June 1988.

⁶ The first station was on 8th June 1989, although the first acoustic interval was only logged after midnight, i.e. on 9th June 1989.

⁷ In cases where a model predicted recruitment is required, the average start date of 2016, 2017 and 2019 (1.267 equating to 8th June) is used.

⁸ In cases where a model predicted recruitment is required, the average cut off length of 2017 (10.5cm) and 2019 (11.5cm) of 11cm is used.

⁹ The first station was on the 5th June 2019, although the first acoustic interval was logged on the 7th June 2019.

Table 3. Sardine total biomass (in tons) as far as Port Alfred and associated CV from the November hydroacoustic survey.

Area	Hondeklop Bay to Cape Point		Cape Point to Cape Agulhas		Hondeklop Bay to Cape Agulhas		Cape Agulhas to Port Alfred	
Year	Sardine Biomass	CV	Sardine Biomass	CV	Sardine Biomass	CV	Sardine Biomass	CV
1984	0	0.000	36453	1.068	36453	1.068	2820	0.509
1985	246	0.514	23188	0.245	23433	0.242	20507	0.767
1986	9494	1.147	228736	1.095	238230	1.052	82690	0.716
1987	76499	0.895	17666	0.489	94165	0.733	19652	0.730
1988	29227	0.663	98817	1.261	128043	0.985	6624	0.514
1989	24223	0.679	174105	0.364	198328	0.330	58327	0.397
1990	6085	2.062	242770	0.386	248855	0.380	42655	0.880
1991	66495	0.558	450685	0.500	517180	0.442	86455	0.624
1992	9417	0.799	238339	0.579	247756	0.557	246710	1.189
1993	93602	1.283	387220	0.509	480822	0.480	79198	0.602
1994	116077	0.882	273653	0.470	389730	0.422	157927	0.725
1995	41943	0.916	322382	0.306	364324	0.291	480402	1.228
1996	24783	0.394	232980	0.379	257763	0.345	271693	0.833
1997	186139	0.791	778696	0.338	964835	0.313	259797	0.981
1998	556200	0.625	526347	0.238	1082547	0.341	524781	0.305
1999	118053	0.710	589976	0.361	708029	0.324	927381	0.280
2000	576865	0.788	149365	0.444	726230	0.633	1698270	0.693
2001	207754	0.725	461863	0.316	669617	0.313	1639983	0.154
2002	14705	0.786	1170007	0.250	1184713	0.247	3072904	0.301
2003	15297	0.577	1327821	0.303	1343118	0.300	2221053	0.258
2004	272	0.720	295836	0.432	296108	0.432	2323193	0.372
2005	146	0.591	75458	0.525	75604	0.524	973386	0.321
2006	61728	1.013	116161	0.335	177889	0.414	534667	0.441
2007	3353	0.526	49785	0.576	53138	0.541	199061	0.421
2008	38	0.753	211833	0.528	211871	0.528	172209	0.682
2009	116051	0.534	146124	0.286	262175	0.285	239400	0.474
2010	105093	0.625	197503	0.398	302597	0.338	198897	0.314
2011	63774	0.297	119051	0.239	182825	0.187	854235	0.283
2012	9881	0.506	176228	0.545	186109	0.517	158945	0.440
2013	2369	0.763	465243	0.434	467613	0.432	144150	0.443
2014	286	0.550	195499	0.476	195786	0.476	248715	0.361
2015	32898	0.565	65569	0.373	98467	0.312	264763	0.391
2016	52469	0.916	130887	0.440	183356	0.409	75219	0.690
2017	3008	0.440	104165	0.356	107173	0.346	227631	0.640
2018	1616	0.735	33229	0.375	34845	0.359	55922	0.783
2019	67	0.789	43553	0.560	43619	0.559	144624	0.652
2020	7091	0.685	44587	0.838	51678	0.729	197263	0.606
2021	-	-	-	-	-	-	-	-
2022	169838	0.442	51654	0.399	221492	0.352	347987	0.578
Feb 2024	269149	0.651	139007	0.374	408156	0.448	587139	0.456

Table 4. Sardine recruitment (in tons and numbers in billions) from Orange River to Cape St. Francis and associated CV from the recruitment acoustic survey. The mean recruit weight is also given (in grams). Blank cells correspond to years/areas for which data are not available, e.g. the survey did not reach Cape St Francis.

Year	Orange River to Cape Point				Cape Point to Cape Agulhas				Orange River to Cape Agulhas				Cape Infanta to Cape St Francis			
	Biomass	CV	Mean Weight	Numbers	Biomass	CV	Mean Weight	Numbers	Biomass	CV	Mean Weight	Numbers	Biomass	CV	Mean Weight	Numbers
1985	19265	0.387	8.168	2.359	17732	1.107	10.663	1.663	36997	0.567	9.200	4.021				
1986	7524	0.743	2.913	2.583	10870	0.629	17.344	0.012	18394	0.480	5.730	2.596				
1987	46506	0.672	9.157	5.079	10297	0.852	19.490	0.761	56803	0.571	10.130	5.840				
1988	1332	0.509	5.936	0.224	22	0.873	6.787	0.003	1354	0.501	5.948	0.228				
1989	4325	0.898	11.320	0.382	3074	0.385	15.695	0.196	7398	0.549	12.803	0.578				
1990	15654	0.803	8.070	1.940	1198	0.779	6.521	0.184	16852	0.748	7.936	2.123				
1991	19166	0.293	10.561	1.815	3	1.133	14.564	0.000	19169	0.293	10.561	1.815				
1992	49245	0.411	9.694	5.080	389	0.429	16.940	0.023	49634	0.407	9.727	5.103				
1993	90812	0.384	6.226	14.585	204	0.565	9.270	0.022	91016	0.383	6.231	14.607				
1994	12742	0.388	9.470	1.346	1375	0.416	19.188	0.072	14117	0.353	9.961	1.417	549	0.467	20.637	0.027
1995	151296	0.429	7.221	20.951	30822	0.537	7.452	4.136	182118	0.368	7.259	25.087	2215	0.491	10.648	0.203
1996	9676	0.313	10.091	0.959	18091	0.618	13.902	1.301	27767	0.417	12.285	2.260				
1997	254638	0.419	7.760	32.816	16122	1.115	13.992	1.152	270761	0.399	7.971	33.968				
1998	102052	0.361	10.582	9.644	1168	0.600	14.329	0.082	103220	0.357	10.613	9.726	3083	0.558	15.037	0.189
1999	29325	0.265	7.540	3.889	34341	0.519	25.883	1.327	63666	0.305	12.206	5.216	76230	0.687	27.188	3.431
2000	196274	0.500	10.614	18.491	7616	0.606	27.609	0.276	203890	0.482	10.864	18.767	95816	0.438	26.472	3.650
2001	553715	0.289	9.268	59.743	3565	0.424	16.151	0.221	557280	0.288	9.294	59.964	1521	0.995	6.844	0.071
2002	316461	0.245	7.786	40.646	107614	0.369	29.575	3.639	424075	0.206	9.576	44.285	117887	0.450	27.550	4.037
2003	150119	0.279	6.231	24.091	224450	0.361	25.716	8.728	374569	0.243	11.413	32.820	84233	0.382	39.799	2.656
2004	14293	0.443	3.956	3.613	11323	0.551	24.517	0.462	25617	0.347	6.287	4.074	4863	0.678	7.688	0.597
2005	11995	0.453	5.054	2.373	2221	0.750	7.494	0.296	14216	0.400	5.325	2.670	11942	0.376	16.835	0.765
2006	24963	0.399	3.178	7.855	15683	0.927	11.455	1.369	40646	0.434	4.407	9.224	80249	0.523	17.310	4.651
2007	17859	0.489	8.515	2.097	14778	0.480	17.827	0.829	32638	0.345	11.153	2.926	19356	0.886	14.429	1.342
2008	10438	0.350	4.326	2.413	11353	0.389	7.191	1.579	21791	0.263	5.459	3.992				
2009	47695	0.895	5.503	8.668	7940	0.360	16.427	0.483	55635	0.769	6.080	9.151	65491	0.987	17.551	3.780
2010	188562	0.500	9.830	19.181	281811	0.731	17.684	15.936	470373	0.481	13.394	35.117	16018	0.664	19.492	0.800
2011	43436	0.563	9.918	4.379	11962	0.408	11.651	1.027	55398	0.450	10.247	5.406				
2012	47043	0.519	9.210	5.108	30638	0.470	18.116	1.691	77681	0.365	11.426	6.799				
2013	81205	0.471	8.147	9.967	14024	0.653	8.128	1.725	95229	0.413	8.145	11.692	27	0.819	13.665	0.002
2014	19575	0.765	11.248	1.740	4059	0.618	21.475	0.189	23634	0.642	12.249	1.929	22327	0.697	21.438	1.045
2015	4357	0.488	13.468	0.324	6530	0.469	29.516	0.221	10887	0.342	19.985	0.545	115900	0.576	22.970	5.370
2016	3989	0.663	22.461	0.178	10390	0.522	14.813	0.701	14379	0.420	16.358	0.879	61053	0.894	26.993	2.264
2017	14330	0.805	2.240	6.397	2036	0.393	9.086	0.224	16365	0.706	2.472	6.621	9471	0.429	11.905	0.818
2018																

Table 4 (continued).

Year	Orange River to Cape Point				Cape Point to Cape Agulhas				Orange River to Cape Agulhas				Cape Infanta to Cape St Francis			
	Biomass	CV	Mean Weight	Numbers	Biomass	CV	Mean Weight	Numbers	Biomass	CV	Mean Weight	Numbers	Biomass	CV	Mean Weight	Numbers
2019	9211	1.200	3.975	2.317	636	0.381	1.315	0.484	9848	1.123	3.516	2.801				
2020	20516	0.594	4.012	5.113	25153	0.603	12.685	1.983	45669	0.426	6.436	7.096				
2021	6985	0.548	3.911	1.786	2434	0.420	7.784	0.313	9419	0.421	4.488	2.099	669	0.427	10.23	0.070
2022	2881	0.334	2.703	1.066	19520	0.744	10.084	1.936	22400	0.650	7.463	3.001	31784	0.407	11.50	2.892
2023	8702	0.645	5.448	1.597	16787	0.657	7.045	2.383	25489	0.485	6.404	3.980	6138	0.884	9.22	0.663

Table 5a. The number of sardine sampled and found infected by length class with the “tetracotyle” type digenean endoparasite, west of Cape Agulhas between 2010 and 2022.

cm	Number sampled												Number infected											
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2022	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2022
7.5	0	5	0	0	0	1	0	0	7	21	0	3	0	0	0	0	0	0	0	0	1	0	0	0
8	2	3	0	0	0	3	1	0	4	31	7	0	0	0	0	0	0	0	0	0	3	0	0	0
8.5	3	0	0	0	1	9	1	0	2	21	4	0	0	0	0	0	0	0	0	0	0	1	0	0
9	0	0	0	0	0	3	4	0	8	4	7	2	0	0	0	0	0	1	0	0	1	1	0	0
9.5	0	0	0	0	0	2	24	0	18	1	5	3	0	0	0	0	0	7	0	0	0	2	0	0
10	3	0	5	0	0	0	30	0	15	3	6	3	0	0	0	0	0	5	0	0	0	0	0	0
10.5	9	0	11	0	2	0	19	0	14	1	12	2	0	0	3	0	0	4	0	0	0	0	0	0
11	7	1	21	0	5	0	25	0	13	5	22	8	0	0	5	0	0	4	0	0	2	1	0	0
11.5	3	2	43	1	5	0	29	0	34	8	21	15	0	0	5	1	1	0	5	0	1	1	3	1
12	16	1	32	4	3	0	33	0	33	15	30	32	0	0	4	3	0	0	6	0	0	2	2	0
12.5	23	4	23	8	5	0	38	0	30	5	38	38	1	1	1	2	1	0	8	0	0	3	1	0
13	35	9	14	25	6	0	44	0	11	11	24	37	1	0	0	7	1	0	6	0	0	4	3	0
13.5	33	11	10	36	12	3	64	0	10	18	24	51	4	0	0	11	2	0	13	0	0	7	2	0
14	40	9	13	22	23	3	42	2	3	16	17	35	5	2	1	8	7	0	7	0	0	6	3	2
14.5	33	3	16	17	23	8	53	2	4	17	14	46	2	0	2	6	2	3	9	0	0	5	3	2
15	30	2	11	8	24	12	43	4	6	17	33	38	6	0	0	5	4	4	15	0	1	6	9	1
15.5	21	8	11	3	8	28	45	8	2	18	30	34	5	3	2	1	3	6	9	3	0	7	7	1
16	24	8	3	0	12	35	29	7	3	15	42	28	9	3	0	0	3	8	12	1	0	6	5	0
16.5	14	22	7	2	13	82	22	6	14	5	35	30	7	3	0	2	4	17	7	3	5	1	12	0
17	9	17	5	4	21	99	24	2	28	2	32	30	5	6	2	0	11	23	10	1	11	0	9	0
17.5	3	73	11	8	43	83	7	1	52	1	26	22	2	17	5	5	23	22	6	0	27	0	8	1
18	12	76	20	10	50	39	18	2	45	1	26	15	1	6	14	10	34	16	11	0	25	1	3	0
18.5	17	98	36	11	68	50	29	4	57	0	13	13	11	24	21	9	51	13	22	2	27	0	3	0
19	17	91	39	6	43	61	59	2	32	1	6	7	9	22	28	5	36	25	43	0	12	1	1	0
19.5	15	36	25	9	49	60	72	2	7	3	0	8	6	5	17	8	41	29	52	0	4	3	0	0
20	3	28	7	0	30	36	42	0	2	1	0	5	3	5	3	0	25	16	32	0	0	0	0	0
20.5	1	7	7	1	21	15	24	1	4	1	1	1	1	2	7	1	19	7	18	0	1	0	1	0
21	0	2	2	0	3	5	2	0	1	1	0	0	0	1	1	0	3	1	2	0	1	0	0	0
21.5	0	1	0	0	1	6	0	0	0	2	0	0	0	0	0	0	1	3	0	0	0	1	0	0
22	0	3	0	0	0	4	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0
22.5	0	5	0	0	0	7	0	0	0	0	0	0	0	3	0	0	0	3	0	0	0	0	0	0
23	0	0	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0

Table 5b. The number of sardine sampled and found infected by length class with the “tetracotyle” type digenean endoparasite, east of 22°E between 2010 and 2022.

cm	Number sampled												Number infected											
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2022	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2022
7.5	1	0	0	0	0	0	5	0	0	8	44	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	3	0	0	0	1	37	0	0	17	43	1	0	0	0	0	0	0	0	0	0	0	0	0
8.5	0	2	0	0	0	2	57	0	1	7	12	1	0	0	0	0	0	0	0	0	1	0	0	0
9	0	5	0	0	3	1	48	0	0	11	2	8	0	0	0	0	0	0	0	0	0	0	0	0
9.5	0	2	0	0	6	3	60	0	1	22	3	10	0	0	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	3	0	18	0	5	39	2	17	0	0	0	0	0	0	0	0	0	0	0	0
10.5	0	0	0	0	5	1	5	0	7	44	1	9	0	0	0	0	0	0	0	0	0	0	0	0
11	0	0	0	0	2	7	2	0	17	11	4	4	0	0	0	0	0	0	0	0	0	0	0	0
11.5	8	0	0	0	5	25	0	0	22	22	3	7	1	0	0	0	0	4	0	0	0	2	2	0
12	18	0	0	0	5	15	2	0	22	19	1	3	1	0	0	0	0	1	0	0	0	1	0	0
12.5	28	1	0	0	3	12	1	1	11	15	1	4	3	0	0	0	0	0	0	0	0	0	0	0
13	31	0	0	0	11	10	0	0	11	5	2	2	2	0	0	0	0	0	0	0	0	0	0	0
13.5	29	1	2	0	15	19	0	0	6	4	1	0	5	0	0	0	0	0	0	0	0	0	0	0
14	11	0	2	0	4	18	1	1	0	1	0	0	3	0	0	0	0	1	0	0	0	0	0	0
14.5	4	2	7	0	10	18	2	6	0	4	8	1	0	1	0	0	0	0	0	0	0	0	0	0
15	0	1	7	1	3	17	0	8	2	24	20	3	0	1	0	0	0	1	0	0	0	0	0	0
15.5	0	2	12	11	0	24	3	25	1	54	45	8	0	1	1	0	0	1	0	0	0	0	0	0
16	0	7	17	9	3	12	0	13	0	38	59	14	0	1	1	1	0	0	0	0	0	0	0	0
16.5	0	26	16	2	5	14	2	14	3	15	72	10	0	2	0	0	0	1	0	0	0	0	2	0
17	0	15	11	1	10	24	2	27	12	3	46	34	0	7	0	0	1	3	0	0	1	0	0	0
17.5	0	44	3	1	12	20	6	30	21	4	29	50	0	12	0	0	1	3	1	0	0	0	0	0
18	0	29	3	0	4	19	10	16	29	0	17	73	0	7	0	0	0	4	0	0	1	0	0	0
18.5	0	72	5	0	7	8	12	2	31	0	7	49	0	24	3	0	1	3	1	0	3	0	1	0
19	0	40	1	0	9	7	6	1	0	0	7	37	0	14	1	0	5	1	1	0	0	0	0	0
19.5	0	44	2	0	14	7	2	2	0	0	1	27	0	18	0	0	11	3	0	0	0	0	0	2
20	0	5	3	0	22	14	4	0	0	0	0	21	0	3	2	0	17	1	1	0	0	0	0	1
20.5	0	2	1	0	20	7	4	0	0	0	1	12	0	0	0	0	16	1	1	0	0	0	0	0
21	0	0	0	0	8	3	0	0	0	0	0	3	0	0	0	0	8	2	0	0	0	0	0	0
21.5	0	1	0	0	3	2	2	0	0	0	0	0	0	1	0	0	2	1	0	0	0	0	0	0
22	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
22.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 5c. The number of sardine sampled and found infected by length class with the “tetracotyle” type digenean endoparasite, east of 20°E between 2010 and 2022.

cm	Number sampled												Number infected											
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2022	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2022
7.5	1	0	0	0	0	0	7	0	0	9	49	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	3	0	0	0	1	38	0	1	37	48	2	0	0	0	0	0	0	0	0	1	0	0	0
8.5	0	2	0	0	0	2	64	0	2	37	19	1	0	0	0	0	0	0	0	0	2	0	0	0
9	1	5	0	0	3	1	59	0	2	18	6	8	0	0	0	0	0	0	0	0	0	1	0	0
9.5	0	2	0	0	7	3	95	0	3	39	6	15	0	0	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	3	0	41	0	18	62	5	24	0	0	0	0	0	0	0	0	0	0	0	0
10.5	0	0	0	0	9	1	10	0	26	62	8	21	0	0	0	0	2	0	0	0	0	0	2	0
11	3	4	0	1	4	7	5	0	51	32	19	16	0	1	0	0	0	0	0	0	2	0	0	0
11.5	18	10	0	2	6	25	2	0	54	51	13	25	1	5	0	0	0	4	0	0	0	3	2	1
12	40	10	0	0	5	15	2	0	62	40	18	32	1	0	0	0	0	1	0	0	0	3	1	1
12.5	51	14	0	11	3	12	1	1	45	28	15	49	3	2	0	1	0	0	0	0	0	0	0	4
13	48	2	0	20	11	10	0	0	25	10	15	36	2	0	0	3	0	0	0	0	0	0	2	2
13.5	50	12	2	41	16	19	0	0	9	9	13	35	6	2	0	5	0	0	0	0	0	0	0	1
14	19	11	2	19	4	18	2	1	2	4	12	8	3	1	0	2	0	1	0	0	0	0	0	2
14.5	6	22	7	5	10	18	4	6	1	8	19	5	0	7	0	0	0	0	0	0	0	1	1	2
15	3	3	7	1	3	17	2	8	2	32	33	6	0	1	0	0	0	1	0	0	0	1	0	0
15.5	3	8	12	11	0	24	10	25	1	61	57	9	0	3	1	0	0	1	1	0	0	2	0	0
16	0	8	17	10	3	12	5	13	0	44	66	15	0	1	1	2	0	0	1	0	0	2	0	0
16.5	0	31	16	3	5	14	8	14	3	21	77	10	0	2	0	0	0	1	1	0	0	1	2	0
17	0	18	11	1	10	24	14	27	12	6	54	34	0	8	0	0	1	3	2	0	1	1	0	0
17.5	0	63	3	3	12	20	12	30	21	4	31	53	0	14	0	1	1	3	1	0	0	0	0	0
18	0	43	3	0	4	19	14	16	30	0	18	77	0	9	0	0	0	4	1	0	1	0	0	0
18.5	0	93	5	0	7	8	13	2	31	0	8	52	0	29	3	0	1	3	1	0	3	0	1	1
19	0	52	1	0	9	7	13	1	0	0	7	39	0	20	1	0	5	1	4	0	0	0	0	0
19.5	0	53	2	0	14	7	16	2	0	0	2	29	0	22	0	0	11	3	4	0	0	0	0	2
20	0	9	3	0	22	14	17	0	0	0	0	21	0	5	2	0	17	1	7	0	0	0	0	1
20.5	0	2	1	0	20	7	12	0	0	0	1	12	0	0	0	0	16	1	5	0	0	0	0	0
21	0	0	0	0	8	3	3	0	0	0	0	3	0	0	0	0	8	2	2	0	0	0	0	0
21.5	0	1	0	0	3	2	6	0	0	0	0	0	0	1	0	0	2	1	2	0	0	0	0	0
22	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
22.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Appendix A: Pelagic sample allocation (repeated from de Moor *et al.* 2024)

The sample allocation method is the process whereby a length frequency (LF) is allocated to every commercial landing, enabling the transformation of the catch to its raised length frequency (RLF). The commercial catch data and field station length frequency data are entered and stored on a Sybase database on the DFFE network. Length frequency data from the Fisheries observer program (2001 to 2011), SAPFIA-funded observer program (2017-presently ongoing) and Industry canneries (2011 to 2018 and from 2020 and presently ongoing) are included in the calculation. The calculations are performed in Access.

Species

It is necessary to calculate LFs for conditioning the assessments for anchovy (*Engraulis encrasicolus*), sardine (*Sardinops sagax*) and round herring (*Etrumeus whiteheadii*); LFs for horse mackerel (*Trachurus trachurus capensis*) are also generated for every run.

Data sources

- Commercial catch: The skipper completes a Pelagic catch Report for every trip and records the estimated catch per set. From 1984 to 2010 this was recorded by a pelagic 10' by 10' block, while from 2011 skippers provided actual latitude and longitudes for each throw as well as the block number position. Every landing is sampled for its species composition and weighed. The fisheries inspectors/scale monitors complete the Pelagic Landing Sheet, based on the actual tonnage offloaded at the landing site. Skipper data (Pelagic Catch Reports) are available on Sybase from 1984 onwards but inspector data (Pelagic landing Sheets) were obtained only from 1987. DFFE field station personnel collect data sheets and enter the information on Sybase.
- Field station samples: DFFE field station personnel collect random samples at the major pelagic fishing harbours for species composition and length frequency (Capricorn Fishing Monitoring was contracted from 2002 until 2005 to man St. Helena Bay and Gansbaai). Samples of industrial fish such as anchovy and round herring are obtained from the top of the hold before the vessel discharges. For this reason industrial samples are obtained mainly from the last throw of the trip. Offloading further damages the already partially-decomposed fish and one cannot sample from the conveyor belt because it would be impossible to weigh those fish. Directed sardine catch, on the other hand, is kept in a very good condition onboard on ice and good quality samples are easily obtained from the conveyor belt, whilst the vessel is discharging. Unfortunately, it is seldom possible to establish which throw is being sampled. Field station data are available on Sybase from 1984 onwards. Ports sampled over the period include Lamberts Bay, Laaiplek, St. Helena Bay, Saldanha, Cape Town, Hout Bay, Kalk Bay, Hermanus, Gansbaai, Mossel Bay and Port Elizabeth. In 2017 it was agreed that a standard seawater allowance of 4.5% for anchovy and round herring and 2.6% for sardine would be implemented (DFFE 2017); this has been implemented since the beginning of the 2018 season.
- Observer samples: The observer program started in 1999 but was terminated in 2011. An industry-funded observer program began in 2017 and is ongoing. Observer samples from 2017 to 2024 have been incorporated into the data thus far. Onboard biological sampling by observers was started in 2001. Observer sampling results reflect an improvement on the field station data because samples are obtained from a known throw, all throws are sampled and the fish is always in a good condition. Unfortunately, the length frequency samples have to be taken ashore for weighing and this gives rise to room for error. The data are stored in an Access database called CAPFISH.

- Industry cannery samples: As part of the quality process at the canneries, fish are also measured and weighed according to DFFE field station specification.

Data extraction from Sybase

- Catch data are extracted from Sybase as text (flat) files; *throw.csv* contains the skippers' data (Pelagic Catch Report) which is proportioned to the actual landed mass and *catch.csv* contains the inspectors' data (Pelagic Landing Sheet).
- Field station data are extracted in the same manner; *spcomp.csv* contains the species composition data and *lfreq.csv* contains the length frequency data.

Data handling and evaluation

DFFE data

- Unfortunately, there is no manual proof reading of all the data, except in cases where the number of throws is excessive (more than 10) and the trip duration is of an unrealistic duration (more than 3 days). Data validation is limited to electronic checking for noticeable mistakes.
- A duplicate dataset of *catch.csv* which is regularly updated by email is kept at Saldanha in an Access table. This means that the data are entered twice, but into separate databases and this allows for the comparison of the two data sets on a regular basis for differences and errors. It might appear unnecessary to keep two data sets, but this is the sole reason that the pelagic catch data remain representative of what was recorded by the scale monitors. The entering of data into Access was stopped since 2015 because these data are now obtained from Industry and used to verify the catches.
- The expected sample weights associated with the length frequency data in *lfreq.csv* are computed and samples that deviate more than 30% are flagged and checked against the raw data. If a flag results from a punch error then the data are corrected, but in the case of a sampling error the record is deleted from the data base.
- Suspect positions, for example areas outside the normal catch areas are checked against the raw data and, if necessary, corrected.

Observer data

- Limited manual proof reading of data.
- Only observer trips that match the commercial data for vessel name and date are used. Mismatched dates do occur, making it very difficult to establish whether a specific vessel carried an observer on a specific date. Therefore, samples from such observer trips are ignored to prevent the inclusion of poor data. Only trips that do link can be used, because the scale monitor's species composition is used to determine the target species of the length frequency sample.
- The structure of the observer length frequency table is altered to make it compatible with the Sybase dataset.
- Only observer length frequencies whose predicted sample weights fall within the set range are used. Data with possible measurement errors or wrong species names are excluded.

Access programs

RLFdata.accdb (where the LFs are generated)

This program has links to the following data sets that are required for the computation.

- 1) Cannery data.accdb (sardine length frequency)
- 2) RSAP reference.accdb (data set with vessel detail and pool positions)
- 3) Capfish.accdb (if data from 2001 to 2011 and from 2017 onwards are included)
- 4) Pel Catch (skipper data sets)

General program outline

- Catches are allocated to pool-area/week strata:
 1. Week: the throw date with the largest catch is used.
 2. Pool area: the existing 21 areas (see Figure A.2) are used, but from 1999 onwards area 21 was subdivided into areas 23 and 24, to accommodate the eastward fishing expansion. The throws within each landing are examined, and the throw with the greatest mass is used as the representative throw.
 3. Assign a target species to every catch. The species with the largest mass is defined as the dominant species in the landing.
- The length frequency samples are grouped by species and target species for the pool-area/week strata and summed.
- A new catch table with additional space for the allocated length frequencies is created.
- The length frequency table is searched and a frequency based on the species, target species, week and pool area criteria are assigned to the catch table.
- In the event of catches not being represented by an appropriate sample, the pool-area/week will be expanded to include surrounding areas and weeks. Stratum expansion continues alternately by week and pool until an appropriate frequency is located.
- If no appropriate sample is found then the average sample for the month is applied. Where no sample for the month exists, the length frequency is estimated using the length frequency of a former month as detailed in the text. Catches of each species and the length frequencies are summed by month over larger user specified areas.
- The length frequencies are exported as Excel files in numbers per length group.

The user specified areas that are used are:

1. Areas 1-6: North of Cape Columbine
2. Areas 7-12: Cape Columbine to Cape Point
3. Areas 13-20: Cape Point to Cape Infanta
4. Area 23: Cape Infanta to Plettenberg Bay
5. Area 24: East of Plettenberg Bay

In 2007, the border between area 23 and 24 was shifted slightly west to 24 degrees east (Tsitsikama), although this made little difference in practice since catches between 23 and 24 were small.

Although the LFs are summarized according to different areas, the allocation process is still based on the original pool areas, with the exception of those cases where pool areas were split by the new borders.

Program changes

In January 2007 four changes were made to the process above:

- The observer length frequencies were included.
- To prevent juvenile sardine frequencies from being allocated to adult sardine catches, the species was separated into directed and bycatch for allocation purposes. This is applicable only when sardine is landed as a bycatch with anchovy. Sardine bycatch with anchovy is mainly juvenile fish whereas bycatch with round herring it is mostly adult fish.
- Noticeable errors in the LF results when the field station catch composition data are used to identify the target species of the length frequency sample, and these composition data differ from those of the scale monitor. Because the field station data are not proofread, and given the inclusion of the observer length frequencies (they also need a target species to be identified), it was decided to standardize on the scale monitors species composition as the only source.
- Missing skipper data (catch area) are catered for. This occurs when the skipper fails to hand in a trip sheet. Currently this is not a major problem but it did happen in the 1980s and 1990s. Where the *catch.csv* file does not have a related record in the *throw.csv* file, the program will search for the most likely catch position, based on the catch type of the other vessels for the same date.

From 2012 the cannery length frequencies are included.

The first change leads to enhanced coverage, especially in the case of industrial fish, i.e. anchovy that are poorly sampled by the field stations. The last three changes were implemented to prevent errors caused by bad data or poor sampling coverage. This can typically be seen in a LF plot as an improbable peak at a certain length group.

In March 2007 an additional change was implemented. Towards the end of the year sporadic landings can be overlooked, because it is not cost effective to continue extensive sampling. These landings are generally small but it is still necessary to allocate a size to the fish. In the past the annual LF average was used, but it was decided to rather allocate the LF from the adjacent month. The LFs are first stratified by area and species type, but where no match is found the requirements for matching area and target species are removed alternatively until a match is found.

Even though throws in multiple pool areas during a single trip do occur, only the catch area for the biggest throw is selected. This is done in order to maintain continuity with the old sample allocation method. A change that could be considered would be to allocate a sample to every throw as opposed to every trip. The scale monitor samples at regular intervals and discrete throws are not sampled. However, if one assumes the species composition of the throws are uniform, then the catch per throw can be calculated, by proportionally applying the species composition to individual throws. Observer sampling is ideally suited for this approach, because every throw is sampled, but greater sampling coverage and matched skipper throws are required.

Sampling coverage

Optimum sample size and sampling coverage can be determined only by using a suitable statistical study, and one can therefore only speculate on the sample size required. Logistical constraints have necessitated a random stratified sampling method, and the grouping of catches and samples on a week/pool-area basis has been adopted since electronic data processing began. Both the sampling and the length frequency approaches are arguably the most suitable considering the fishing strategy and the available data. The percentage coverage per stratum is readily quantified, and the first level pool-area/week coverage could possibly be used as an index of sampling coverage. 100 percent coverage is not attainable because

of financial and logistical constraints, and it is more than likely unnecessary. From Figure A.1 it appears that 80 percent coverage is attainable.

Many factors influence the relationship between the number of samples taken and the coverage obtained, but in general more samples will lead to better coverage. In earlier years, field stations were well manned and more samples were taken than presently. The Observer program was introduced in 2000 and this improved sampling and as a result field station sampling was reduced. However, when the program was terminated in 2011 the field station coverage was not adjusted to earlier levels. This poor coverage can be seen as the low values on the graph in Figure A.1. The inclusion of the industry-funded observer samples since 2017 have assisted coverage in recent years (Coetzee and Merkle 2019). The fisheries cannery data (used since 2012) is necessary to maintain directed sardine sampling levels.

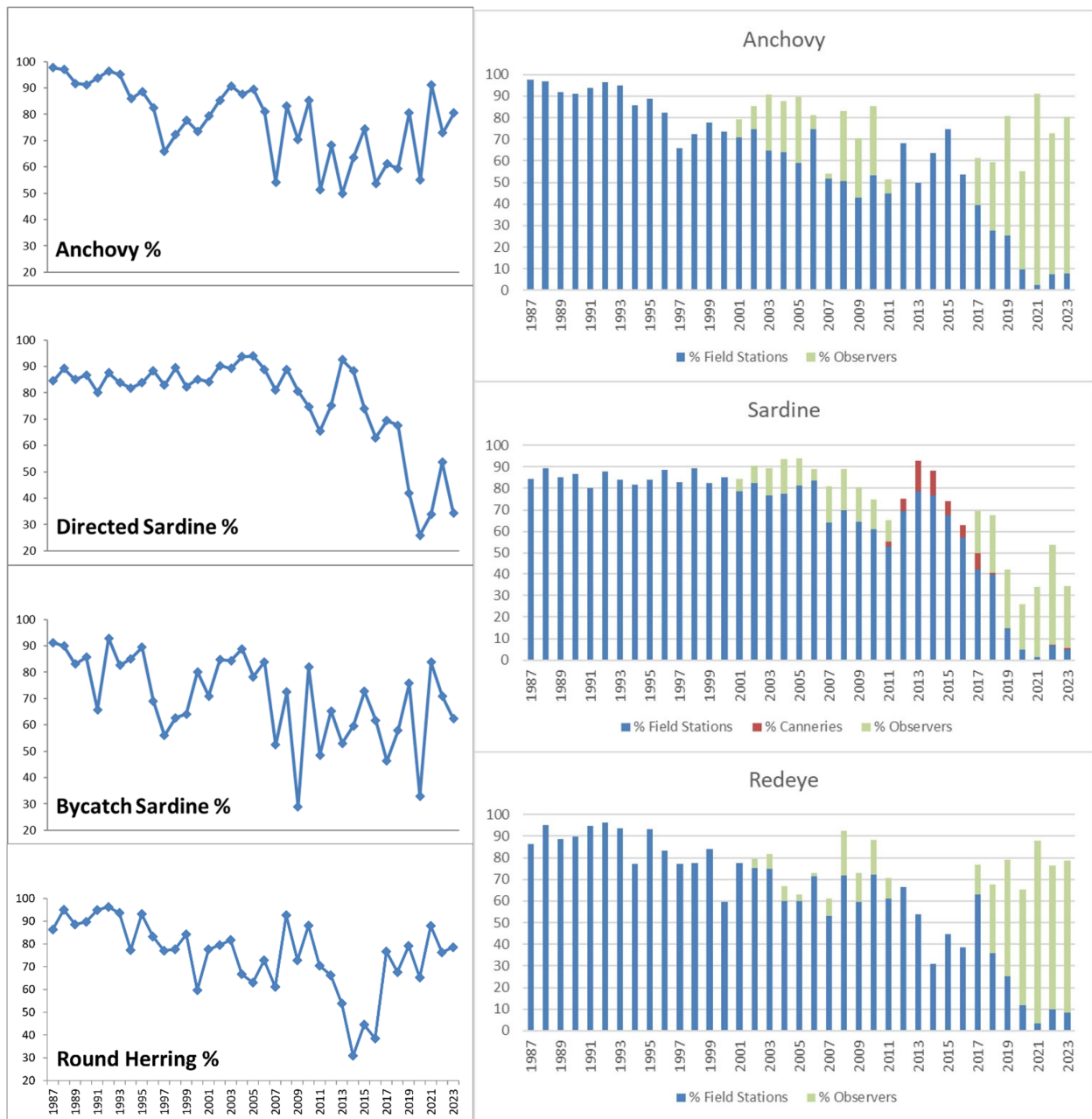


Figure A.1. Sampling coverage obtained on a first level pool-area/week including 2000-2011 and 2017 to 2023 observer data. The histograms indicate the proportion of sampling obtained from observer data.

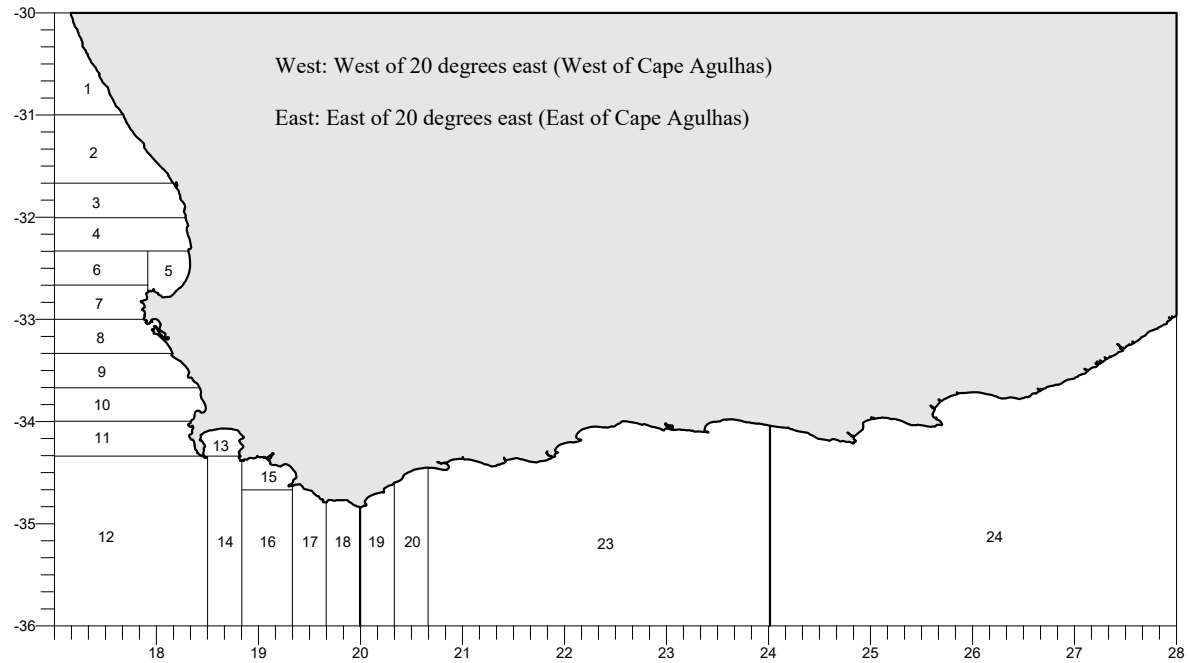


Figure A.2. The pool areas that are used for sample allocation.

Appendix B: Method Used to Calculate Recruit Biomass (repeated from de Moor *et al.* 2024)

To estimate recruit-only biomass and CVs (rather than that of recruits and adults combined) the acoustic density is apportioned between recruits and adults for each interval, based on the length- specific target strength and a cut-off length (derived independently from maturity at length information). Essentially, the density in each interval is multiplied by the proportion of recruits in that interval to get a recruit only density. The proportion of recruits in each interval is obtained by calculating the proportion of acoustic energy backscattered by recruits only, based on the length frequency that each interval has been assigned and a cut-off length:

- 1) For each trawl (grid) the acoustic back scattering for each length class is calculated for each species and multiplied by the number of fish in that length class (basically applying the species specific target strength relationship to the length class (L_t)):

$$BS = \begin{cases} 10^{0.1 \times -21.12} \times L_t^{-12.15/10} \times N & \text{if } Sp = 1 \\ 10^{0.1 \times -13.21} \times L_t^{-14.9/10} \times N & \text{if } Sp = 2 \text{ or } 5 \\ 10^{0.1 \times -7.75} \times L_t^{-15.44/10} \times N & \text{if } Sp = 3 \text{ or } 4 \end{cases}$$

where $Sp\ 1$ = anchovy, $Sp\ 2$ = sardine, $Sp\ 3$ = horse mackerel, $Sp\ 4$ = mackerel and $Sp\ 5$ = round herring.

- 2) The backscattering (BS) is summed for each species for each trawl to give a total backscatter for each grid, BS_{tot} .
- 3) The backscattering due to recruits, BS_{rec} , is then calculating by summing BS for only the length classes less than the cut-off length for each species for each trawl. The cut-off length is obtained from the length at 20% maturity, derived from samples collected during the survey (Coetzee *et al.* 2018, Yemane *et al.* 2024).
- 4) The proportion of recruits in each trawl is then calculated by BS_{rec}/BS_{tot} .
- 5) This proportion is then multiplied by the original interval density (of recruits and adults) to obtain the recruit only density per interval, which is then used to estimate the biomass and CV of recruits following the Jolly and Hampton (1990) method.
- 6) This recruit only density is used in the regressions¹⁰ of capped to uncapped data in order to estimate (using the Jolly and Hampton weighting procedure) the uncapped recruit only biomass prior to 1997 together with a CV.

A weighted length frequency is then derived as follows:

- 7) Each acoustic interval is linked to a particular grid reference (trawl sample) which was used to scale the acoustic energy to density. The trawl sample has a length frequency (LF) and associated length frequency mass (LFMASS). This LF and LFMASS include both adults and recruits.
- 8) For each interval, the acoustic density is multiplied by the interval length. This weighted interval density is then summed over all intervals for each grid reference, per stratum and per species to give an acoustic weighting to each grid reference, $w_{GR}(grid, stratum, species)$.
- 9) The weighted grid reference is then summed over all grid references for each stratum and species to give a weighted grid reference per stratum for each species, $w_{GR}(stratum, species)$.
- 10) For each length class of each grid reference, a Trawl WF (trawl weighting factor) $= w_{GR}(grid, stratum, species)/LFMASS$ is calculated. This converts the acoustic weighting (in terms of mass) into a factor in terms of numbers.
- 11) The length frequency (LF) is then weighted by this Trawl WF and summed for each length class to give a weighting to each length class ($Lgroup$) for each stratum for each species $sum(number \times trawl\ WF)$, $WLF(Lgroup, stratum, species)$.

¹⁰ The regressions were, however, calibrated with total (recruit and adult) densities (de Moor *et al.* 2008).

- 12) $WLF(Lgroup, stratum, species)$ is then scaled to the biomass of the stratum: $BLF(Lgroup, stratum, species) = [WLF(Lgroup, stratum, species) * Biomass(stratum, species)] / \sum w_{GR}(stratum, species)$.
- 13) BLF is then summed across all strata for each species to give a final length frequency per species for the survey (this is done separately up to Cape Infanta and for the whole survey).

Appendix C: Cut-off lengths for ≤ 14 cm bycatches at the time of the November survey

The following figures show the annual November survey length frequencies together with the cut-off length to be applied for 0-year olds in the commercial bycatch in November (grey dotted line) based on the recruit survey cut-off length in the subsequent year less 6cm, and the cut-off length that would be applied for 0-year olds in the commercial bycatch in September of the same year as the survey (black dotted line) based on the recruit survey cut-off length of the same year + 1.5cm. Note that the y-axis scales are not the same for all figures, as the scale has been adjusted to assist visibility. There was no November survey in 2021.

The cut-off length to be applied to commercial bycatch for 0-year-olds in November typically results in very few 0-year olds being *relatively* present in the November survey, as expected. Exceptions include Novembers 1984, 1998, 2006, 2008, 2014, 2018, 2023. The cut-off length to be applied to commercial bycatch for 0-year-olds in September frequently includes at least one mode in the survey distribution, as expected, but the proportion varies substantially between years. This is not unexpected for a short-lived species. The minimum September 0-year-old cut-off length is 10.5cm corresponding to a recruit survey cut-off-length of 9cm in 1988, with the next lowest being 12cm (10.5cm in May 2017). The maximum September 0-year-old cut-off length is 17.0cm; this is within 2cm of the length at 50% maturity (Yemane et al. 2024), except in 1989, 1992, 1995. 2009, 2011, 2012, 2014, 2015, 2016 where the length at 50% maturity was 2.5-4.5cm less.

